

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279537

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

WATANABE; Masayoshi et al.

ELECTRICITY-DRIVEN VEHICLE

Abstract

An electricity-driven vehicle includes: a floor panel; a depressed portion of a groove shape, that is formed on the floor panel and that extends in a vehicle width direction; a battery casing that is placed in the depressed portion; one or more batteries of a replaceable type, that can be attached to and detached from the battery casing; a smoke exhaust duct that extends from the battery casing, that penetrates through a bottom surface of the depressed portion, and that is opened to an outside of the vehicle; and a check valve that is provided on the smoke exhaust duct, and that prohibits inflow of a fluid from the outside of the vehicle into the battery casing.

Inventors: WATANABE; Masayoshi (Nagoya-shi, JP), SAITO; Kan (Ikeda-shi, JP), TSUJITA; Toshiya (Ikeda-shi, JP), NAKAMURA; Naohiro (Ikeda-shi, JP), KAWAMOTO; Yasuo (Ikeda-shi, JP), OGAMI; Hideki (Ikeda-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Aichi-ken, JP); ISUZU MOTORS LIMITED (Kanagawa, JP)

Family ID: 96829444

Appl. No.: 19/062073

Filed: February 25, 2025

Foreign Application Priority Data

JP	2024-029866	Feb. 29, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H01M50/358 (20210101); B60K1/04 (20190101); H01M10/658 (20140101); H01M50/249 (20210101); H01M50/317 (20210101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-029866 filed on Feb. 29, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

[0002] The present disclosure relates to an electricity-driven vehicle, equipped with a replaceable battery.

BACKGROUND

[0003] In the related art, an electricity-driven vehicle is widely known, in which a motor which is a motive power source, and a battery which supplies electric power to the motor are provided on a vehicle. The battery equipped in the electricity-driven vehicle is a secondary battery which can be charged and discharged, and is charged by an external power source as necessary.

[0004] A configuration is being proposed which employs a replaceable battery, which can be attached to and detached from the electricity-driven vehicle, in order to effectively utilize the electricity-driven vehicle even during charging of the battery. In a structure with a replaceable battery, when a state of charge of the battery equipped on the electricity-driven vehicle becomes low, the battery is removed from the electricity-driven vehicle, and is replaced with another battery which is charged in advance. The battery which is removed from the electricity-driven vehicle is then charged outside the electricity-driven vehicle. Because the electricity-driven vehicle can travel even during the charging of the removed battery, the electricity-driven vehicle can be more effectively utilized.

[0005] In such a secondary battery, it is known that gas is generated due to internal short-circuiting, external short-circuiting, or the like. JP 2017-091824 A discloses a battery unit having a release passage through which such gas flows. According to the technique of JP 2017-091824 A, the gas generated from the battery is guided to the outside of a casing of the battery unit, without remaining in the casing.

[0006] When the battery is to be placed in a space which is not completely separated from a vehicle cabin, a structure is demanded which prevents inflow of the gas into the vehicle cabin. However, JP 2017-091824 A does not consider processing of gas guided to the outside of the casing.

[0007] An advantage of the present disclosure lies in an electricity-driven vehicle which can prevent inflow of gas generated from a battery into a vehicle cabin.

SUMMARY

[0008] According to one aspect of the present disclosure, there is provided an electricity-driven vehicle comprising: a floor panel; a depressed portion of a groove shape, that is formed on the floor panel and that extends in a vehicle width direction; a battery casing that is placed in the depressed portion; one or more batteries of a replaceable type, that can be attached to and detached from the battery casing; a smoke exhaust duct that extends from the battery casing, that penetrates through a bottom surface of the depressed portion, and that is opened to an outside of the vehicle; and a check valve that is provided on the smoke exhaust duct, and that prohibits inflow of a fluid from the outside of the vehicle into the battery casing.

[0009] With the above-described structure, the gas generated in the battery can be released to the outside of the vehicle through the smoke exhaust duct. As a result, the inflow of the gas generated from the battery into the vehicle cabin can be prevented.

[0010] In this case, the electricity-driven vehicle may further comprise a deck board that covers and hides the depressed portion; and a thermal insulation member placed between the deck board and the battery casing.

[0011] With the above-described structure, transfer of heat generated in the battery to an upper space of the deck board can be prevented. As a result, degradation of baggage placed in the vehicle cabin, and discomfort of passengers in the vehicle cabin can be reduced.

[0012] Further, a loading port of the battery may be formed at one end of the battery casing in the vehicle width direction, and the smoke exhaust duct may extend from an end surface of the battery casing at a side opposite the loading port.

[0013] With the above-described structure, a terminating end of the smoke exhaust duct is not hidden by the battery casing. Thus, an operator can assemble the battery casing while visually checking the position of the terminating end of the smoke exhaust duct. As a result, the assembling process of the battery casing can be simplified.

[0014] The smoke exhaust duct may penetrate through the bottom surface of the depressed portion approximately at a center in the vehicle width direction.

[0015] With the above-described structure, even when the battery casing is placed with left and right sides inverted, the position of the hole through which the smoke exhaust duct penetrates does not need to be changed.

[0016] The smoke exhaust duct may extend from a bottom surface or a vehicle-rear-end surface of the battery casing.

[0017] By providing the smoke exhaust duct at the bottom surface of the battery casing, a dimension of the battery casing in the vehicle width direction other than the smoke exhaust duct can be increased. Further, by providing the smoke exhaust duct at the vehicle-rear-end surface of the battery casing, the smoke exhaust duct can be separated from the front seat (and, consequently, the passengers).

[0018] According to an aspect of the present disclosure, inflow of gas generated from the battery into the vehicle cabin can be prevented.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] Embodiment(s) of the present disclosure will be described based on the following figures, wherein:

[0020] FIG. 1 is a perspective diagram showing a region near a rear side door of an electricity-driven vehicle, viewed from outside the vehicle;

[0021] FIG. 2 is a schematic plan view of the electricity-driven vehicle;

[0022] FIG. 3 is a cross-sectional diagram along an A-A line of FIG. 2;

[0023] FIG. 4 is a cross-sectional diagram along a B-B line of FIG. 2;

[0024] FIG. 5 is a diagram showing another example placement of a smoke exhaust duct; and

[0025] FIG. 6 is a diagram showing yet another example placement of the smoke exhaust duct.

DESCRIPTION OF EMBODIMENTS

[0026] A structure of an electricity-driven vehicle **10** according to an embodiment of the present disclosure will now be described with reference to the drawings. FIG. 1 is a perspective diagram showing a region near a rear side door **12** of the electricity-driven vehicle **10**, viewed from outside the vehicle. FIG. 2 is a schematic plan view of the electricity-driven vehicle **10**. FIG. 3 is a cross-sectional diagram along an A-A line of FIG. 2, and FIG. 4 is a cross-sectional diagram along a B-B line of FIG. 2. In the drawings, “Fr”, “Up”, and “Rh” respectively refer to a front side, an upper side, and a right side of the electricity-driven vehicle **10**.

[0027] The electricity-driven vehicle **10** comprises a traveling motor (not shown), and a battery **44**

which supplies electric power to the traveling motor. The electricity-driven vehicle **10** exemplified below is a commercial vehicle for transporting baggage. A rear seat of the electricity-driven vehicle **10** is removed, in order to improve a loading efficiency of the baggage. Therefore, an entirety of a rear side of a front seat **29** (not shown in FIG. **1**; refer to FIG. **3**) can be used as a baggage compartment. On a side of the electricity-driven vehicle **10**, a front side door **11** and a rear side door **12** are placed, aligned in a front-and-rear direction. As shown in FIG. **2**, the front side door **11** is a hinge door which swings about an axis extending in a vertical direction, and the rear side door **12** is a sliding door which slides in a vehicle front-and-rear direction. The rear side door **12** opens and closes a rear side door opening **14**.

[0028] The battery **44** is a replaceable battery which can be easily replaced in the electricity-driven vehicle **10**. The battery **44** is formed by housing a battery body formed from a plurality of unit cells in a housing. The housing of the battery **44** has, for example, a parallelepiped shape elongated in a vehicle width direction. A battery-side connector is provided at an end of the battery **44**. When the battery-side connector is connected to a vehicle-side connector, an electric power system equipped on the vehicle and the battery **44** are electrically connected with each other. The battery body is a secondary battery (for example, a lithium ion battery or the like) which can be charged and discharged. Therefore, the battery **44** which is removed from the electricity-driven vehicle **10** can be charged externally to the vehicle.

[0029] By setting the battery **44** as a replaceable battery in this manner, the usage efficiency of the electricity-driven vehicle **10** can be improved, and a demand peak of commercial electric power can be reduced. That is, when the battery **44** is of the replaceable type, after the battery **44** is removed from the electricity-driven vehicle **10**, another battery **44** which is already charged can be mounted on the electricity-driven vehicle **10**. Further, with such a configuration, because the electricity-driven vehicle **10** can travel even during a period in which a part of the battery **44** is charged, the usage efficiency of the electricity-driven vehicle **10** is improved. In addition, when the battery **44** is of the replaceable type, time of performing the charging operation can be freely selected, regardless of a usage situation of the electricity-driven vehicle **10**. Because of this, the battery **44** can be charged during a time period of low demand of the commercial electric power (for example, late at night), and the demand peak of the commercial electric power can be reduced.

[0030] Such a battery **44** is mounted at a location where a user can easily replace the battery and where loading of the baggage is not obstructed. In the case of the electricity-driven vehicle **10** of the present disclosure, a battery loading portion **40** at which the battery **44** is loaded is provided at an adjacent position of the rear side door opening **14**. The battery loading portion **40** will now be described in detail.

[0031] First, prior to the description of the battery loading portion **40**, a basic vehicle body structure of the electricity-driven vehicle **10** will be described. The electricity-driven vehicle **10** has a vehicle body structure approximately identical to that of existing vehicles. By using the vehicle body structure of the existing vehicles as is, components and manufacturing processes can be made common with the existing vehicles, resulting in reduction of development and manufacturing costs of the electricity-driven vehicle **10**. As shown in FIGS. **2** and **4**, the electricity-driven vehicle **10** comprises a pair of side members **24** placed with a spacing in the vehicle width direction, and a plurality of cross members **26** connecting the pair of side members **24**. Each of the side members **24** and the cross members **26** is a frame of the electricity-driven vehicle **10**.

[0032] As shown in FIG. **4**, at an upper side of the side member **24**, a floor panel **16** is provided. The floor panel **16** is a steel plate panel forming a floor surface of a vehicle cabin. The floor panel **16** is welded to the frame (side member **24** or the like) of the electricity-driven vehicle **10**. As shown in FIG. **3**, at an immediately rear side of the front seat **29**, a depressed portion **18** is present. The depressed portion **18** is a portion which extends in the vehicle width direction and which is depressed in a groove shape. In the case of the existing vehicles, a rear seat is placed at an immediately rear side of the depressed portion **18**. As described above, in the case of the electricity-

driven vehicle **10** of the present embodiment, the rear seat is removed.

[0033] At a laterally adjacent position of the depressed portion **18**, the rear side door opening **14** is placed. As shown in FIG. **4**, a lower end of the rear side door opening **14** is defined by a rocker **28**. The rocker **28** is one type of a frame extending in the vehicle front-and-rear direction. In the present embodiment, as shown in FIG. **4**, the battery loading portion **40** is provided in the depressed portion **18**. In addition, as shown in FIG. **2**, the battery loading portion **40** is completely fitted within a width of the rear side door opening **14** in the front-and-rear direction. By employing such a configuration, the user can easily access the battery loading portion **40** through the rear side door opening **14**.

[0034] The battery loading portion **40** is a part at which the battery **44** is loaded. The battery loading portion **40** has a battery casing **42**. As shown in FIG. **4**, the battery casing **42** is a container which houses one or more batteries **44** (in the illustrated example configuration, three batteries **44**), and is a box approximately having a parallelepiped shape with one end in the vehicle width direction opened. The opening of the battery casing **44** at one end in the vehicle width direction serves as a loading port **48** through which the battery **44** is inserted and removed. When the battery **44** is to be replaced, the user accesses the battery casing **42** from a side of the electricity-driven vehicle **10** (in the illustrated example configuration, the left side), and inserts or removes the battery **44** via the loading port **48**. As described above, in the present embodiment, the rear side door **12** is the sliding door. By employing the sliding door, interference between the battery **44** to be inserted or removed and the rear side door **12** can be effectively prevented, and workability when the battery **44** is replaced can be improved.

[0035] The loading port **48** is opened and closed by a casing lid **50**. The casing lid **50** airtightly covers the loading port **48**. In order to maintain airtightness, for example, on at least one member of the casing lid **50** and the loading port **48**, a sealing member **51** which airtightly contacts the other member is attached. By providing the airtightness to the casing lid **50** in this manner, it becomes possible to prevent inflow of the gas to be described later into the vehicle cabin. The handling of the gas will be described later.

[0036] In addition, in the case of the present embodiment, the casing lid **50** swings with the center near a lower end of the loading port **48**. As shown in FIG. **4**, the casing lid **50** closes the loading port **48** when in a standing orientation approximately parallel with a perpendicular surface. When the casing lid **50** in the standing orientation swings downward, the loading port **48** is opened. By swinging the casing lid **50** about the lower end of the loading port **48** in this manner, a configuration is achieved in which, when the casing lid **50** is opened, the casing lid **50** does not interpose between eyes of the user (which are normally positioned above the loading port **48**) and the loading port **48**. With this configuration, the user can more easily see the periphery of the loading port **48**, and, as a consequence, the workability when the battery **44** is replaced can be improved. In addition, in the case of the illustrated example configuration, a dimension of the casing lid **50** in the up-and-down direction is significantly smaller than the dimension of the casing lid **50** in the front-and-rear direction. Thus, by employing a configuration where the casing lid **50** opens downward, as compared with a case where the casing lid **50** opens laterally, an amount of protrusion when the casing lid **50** is opened can be set small. However, the structure of the casing lid **50** is not limited to the illustrated example configuration, and may be suitably changed. For example, the casing lid **50** may be changed to a laterally opening type, a double door type, or a sliding type.

[0037] As shown in FIGS. **2** and **4**, the battery loading portion **40** (that is, the battery casing **42**) is fitted within a width between the two side members **24** in the plan view. With such a configuration, when the electricity-driven vehicle **10** collides at the side, the side member **24** receives the impact before the battery loading portion **40**. Because of this, damage to the battery **44** housed in the battery loading portion **40** is effectively prevented.

[0038] As shown in FIG. **3**, a height of the battery casing **42** is shorter than a depth of the depressed

portion **18**. The electricity-driven vehicle **10** further comprises a deck board **30** placed at an upper side of the floor panel **16** so as to cover and hide the depressed portion **18**. In other words, the battery loading portion **40** is placed in a space surrounded by the deck board **30** and the depressed portion **18**. As shown in FIGS. **2** and **3**, the deck board **30** forms a flat floor surface, extending from a rear side of the front seat **29** to a rear end of the vehicle. By providing such a deck board **30**, the entirety of the space behind the front seat **29** can be used as a baggage compartment, and thus, a transport efficiency of the baggage by the electricity-driven vehicle **10** can be improved.

[0039] The deck board **30** is not welded to any of the members of the electricity-driven vehicle **10**, and can be removed from the electricity-driven vehicle **10** without destroying a component of the electricity-driven vehicle **10**. By removing the deck board **30** from the electricity-driven vehicle **10** as necessary, maintenance or the like of the battery loading portion **40** can be easily performed.

[0040] As described above, a baggage or the like is placed at an upper side of the deck board **30**. In the present embodiment, a thermal insulation member **60** is placed between the deck board **30** and the battery casing **42**. The thermal insulation member **60** may be attached to the deck board **30**, or attached to a top surface of the battery casing **42**. Further, the thermal insulation member **60** may be a separate member, separated from both the deck board **30** and the battery casing **42**.

[0041] In either case, by providing the thermal insulation member **60**, transfer of heat of the battery **44** to an upper side of the deck board **30** (and consequently, the baggage) can be prevented. With this configuration, degradation of the baggage, and discomfort of the passengers due to an increase in the temperature can be suppressed.

[0042] Here, gas may be released from the battery **44** due to overcharging, overdischarging, short-circuiting, or the like. The replaceable battery **44** is configured such that the gas can be released to the outside of the battery **44**. The gas has a high temperature, and in some cases, an odor. If such gas flows into the space at the upper side of the deck board **30**, degradation of the baggage or discomfort of the passengers may result.

[0043] In consideration of this, in the present embodiment, a smoke exhaust duct **52** for guiding the gas to outside the vehicle is provided. As shown in FIG. **4**, the smoke exhaust duct **52** is a duct which extends from an end surface of the battery casing **42** at a side opposite the loading port **48**, penetrates through a bottom surface of the depressed portion **18**, and is opened to the outside of the vehicle. The gas generated from the battery **44** is released to the outside of the vehicle via the smoke exhaust duct **52**. In this manner, by releasing the gas to a lower side of the floor panel **16**, inflow of the gas into the upper space of the deck board **30** can be effectively prevented.

[0044] A check valve **54** is attached to the smoke exhaust duct **52**. The check valve **54** permits passage of a fluid from the battery casing **42** to the outside of the vehicle, but prohibits passage of the fluid from the outside of the vehicle to the battery casing **42**. By providing the check valve **54**, a back flow of the gas released to the outside of the vehicle into the battery casing **42** can be prevented.

[0045] Here, obviously, a through hole through which the smoke exhaust duct **52** is inserted is formed on the bottom surface of the depressed portion **18**. In the example configuration of FIG. **4**, the smoke exhaust duct **52** extends from the end surface of the battery casing **42** in the vehicle width direction, and the through hole of the depressed portion **18** is positioned at an outer side of the battery casing **42**. With such a placement, when assembling the battery casing **42**, the operator can easily view the position of the through hole of the depressed portion **18**. As a result, the assembling operation of the battery casing **42** can be simplified.

[0046] However, the placement of the smoke exhaust duct **52** is not limited to the example configuration of FIG. **4**, and may be suitably changed. For example, as shown in FIG. **5**, the smoke exhaust duct **52** may extend from a bottom surface of the battery casing **42**. With such a configuration, a dimension of the battery casing **42** in the vehicle width direction can be increased as compared with the example configuration of FIG. **4**.

[0047] In addition, as shown in FIG. **5**, the smoke exhaust duct **52** may penetrate through the

bottom surface of the depressed portion **18** approximately at the center in the vehicle width direction. With this configuration, an extent of a structural change of the electricity-driven vehicle **10** when left and right sides of the battery loading portion **40** are inverted can be reduced. Specifically, in FIG. 5, a configuration is employed in which the battery **44** is replaced from the left side of the electricity-driven vehicle **10**. However, in some cases, depending on the user's desires, it may be desired to replace the battery **44** from the right side of the electricity-driven vehicle **10**. In order to fulfill such desires of the user, the placement of the battery loading portion **40** may in some cases be inverted in the left and right sides, before the user purchases the electricity-driven vehicle **10**. In the case of FIG. 4, if the placement of the battery loading portion **40** is inverted in regard to the left and right sides, the position of the through hole into which the smoke exhaust duct **52** is inserted significantly changes. On the other hand, if the smoke exhaust duct **52** is provided approximately at the center in the vehicle width direction as shown in FIG. 5, the position of the through hole does not change even when the placement of the battery loading portion **40** is inverted in the left and right sides. As a result, the extent of the structural change of the electricity-driven vehicle **10** due to the left-and-right inversion of the battery loading portion **40** can be set small. [0048] As another form, as shown in FIG. 6, the smoke exhaust duct **52** may extend from a rear end surface of the battery casing **42**. With such a configuration, the front seat **29** (and consequently, the passengers) can be separated from the flow path of the gas. As a result, the flow of the gas to the periphery of the passengers can be more reliably prevented. In this case also, the smoke exhaust duct **52** may be placed approximately at the center in the vehicle width direction. [0049] The structures described above are merely exemplary, and, so long as the electricity-driven vehicle has the structure described in claim **1**, the other structures may be suitably changed. For example, in the above description, the rear side door **12** is described as a sliding door, but alternatively, the rear side door **12** may be a hinge door, similar to the front side door **11**. Further, the structure of the battery loading portion **40** may also be suitably changed. For example, the shape and the size of the battery loading portion **40** may be suitably changed. Moreover, the number and placement of the smoke exhaust duct **52** may be suitably changed. For example, a plurality of the smoke exhaust ducts **52** may be provided.

REFERENCE SIGNS LIST

[0050] **10** electricity-driven vehicle, **11** front side door, **12** rear side door, **14** rear side door opening, **16** floor panel, **18** depressed portion, **24** side member, **26** cross member, **28** rocker, **29** front seat, **30** deck board, **40** battery loading portion, **42** battery casing, **44** battery, **48** loading port, **50** casing lid, **51** sealing member, **52** smoke exhaust duct, **54** check valve, **60** thermal insulation member

Claims

1. An electricity-driven vehicle comprising: a floor panel; a depressed portion of a groove shape, that is formed on the floor panel and that extends in a vehicle width direction; a battery casing that is placed in the depressed portion; one or more batteries of a replaceable type, that can be attached to and detached from the battery casing; a smoke exhaust duct that extends from the battery casing, that penetrates through a bottom surface of the depressed portion, and that is opened to an outside of the vehicle; and a check valve that is provided on the smoke exhaust duct, and that prohibits inflow of a fluid from the outside of the vehicle into the battery casing.
2. The electricity-driven vehicle according to claim 1, further comprising: a deck board that covers and hides the depressed portion; and a thermal insulation member placed between the deck board and the battery casing.
3. The electricity-driven vehicle according to claim 1, wherein a loading port of the battery is formed at one end of the battery casing in the vehicle width direction, and the smoke exhaust duct extends from an end surface of the battery casing at a side opposite the loading port.
4. The electricity-driven vehicle according to claim 1, wherein the smoke exhaust duct penetrates

through the bottom surface of the depressed portion approximately at a center in the vehicle width direction.

5. The electricity-driven vehicle according to claim 1, wherein the smoke exhaust duct extends from a bottom surface or a vehicle-rear-end surface of the battery casing.
